Not at a distance, but nowhere: non-locality beyond entanglement, in twelve phenomena

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Abstract

In physics «non-locality» mostly names one fact: the correlations of entanglement. In Mathematical Materiality (MM) it is the foundation: a non-local phase, not located in space and time and present everywhere, of which the local world is the count made from a point. Here «non-local» means not located in space and time; it does not mean action at a distance. This record collects, from MM's published records and the author's premises, twelve phenomena in which MM's reading uses the same non-local phase as the reason, and shows that entanglement is the case in which it rests most directly on a measured datum, not the only one. There are no new calculations: the result is the unity of the reading. The twelve phenomena:

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How to read this text. Every technical word is made clear the first time it appears. You will find it again in the glossary, at the end. Numbers such as 2.48 or 2.68 indicate other works by the author, published on the Zenodo archive: we call them *deposits*. The list with the DOIs is in the section Deposits cited, at the end. How the work was done (checks, reviews, files) is told in the section How the work was done, before the glossary.

In short

Mathematical Materiality (MM) is the author's framework of ideas about physics. MM starts from a **non-local phase**. It is reality seen as relations that are not in a place or at an instant. This phase is present everywhere. The **local** world, the world of places and instants, is the same reality counted from a point.

Usually non-locality makes one think of **entanglement**. Entanglement is a state of two particles, even distant ones, that must be described together: the results of the measurements on one and on the other are linked. In entanglement the non-local phase rests more directly on a measured datum. But entanglement is not the only case.

This deposit collects twelve phenomena. In all of them MM's reading uses the same non-local phase as the reason. Here they are:

- the correlations of entanglement and their ceiling, that is, the highest value they reach;
- the laws of physics, the same everywhere;
- identical particles, such as two electrons;
- the objects that two observers see alike;
- the boundary of the local world;

- probability;
- the threshold beyond which an object stops being at one point;
- the present;
- space and time themselves;
- the limit of light;
- dark matter;
- the boundary of a gravitating system, that is, of a system held together by gravity.

For each phenomenon the deposit says six things:

- what is observed;
- how known physics reads it;
- what the non-local phase does;
- what rests on calculations already deposited;
- what remains open;
- where the link comes from: from a deposit, from a premise of the author or from a proposal of the AI.

A *premise* is a starting sentence of the author: a fixed point, like a postulate. The *AI* is the artificial intelligence that wrote the text at the author's request.

There are no new calculations. The result is the unity of the reading: the same idea works in all twelve phenomena.

The question

What physics usually means

In physics the word "non-locality" mostly names one fact: the correlations of entanglement.

Two results are *correlated* when they agree more often than chance would. An example with everyday objects. Take a pair of gloves. Put one glove in a box and the other in another box, without looking. Send the two boxes to two distant cities. Whoever opens the first box and finds the right glove knows at once that the left one is in the other. No message has gone from one box to the other. The information was in the boxes from the start.

In 1964 the physicist John Bell proved a theorem on the correlations between two distant systems. It is **Bell's theorem**. It starts from three assumptions:

- each system carries its own properties with it, like the gloves in the boxes;
- no influence travels faster than light;
- the choices of the measurements are independent of the state of the source, that is, of what produces the two systems.

The theorem shows that these three assumptions, together, put a limit on the correlations. It is the **Bell limit**. The theorem is used in this form also in deposit 2.68.

Experiments exceed that limit. And physics treats non-locality as a property of *entangled* systems, that is, systems linked by entanglement.

The rest of physics is local

Elsewhere physics is local: it describes each thing in a place and at an instant. Physics accounts for some very general facts with separate assumptions, one for each fact:

- **the laws are the same everywhere.** Physics accounts for this with the *homogeneity* of space and time: the assumption that no place and no instant is special;
- **all electrons are identical.** Physics accounts for this with a *field* for each species of particle (we will see what a field is in §3.3);
- **two observers see the same object.** Physics accounts for this with the *redundancy* of the copies in the environment, that is, with the many traces of the object that spread around (§3.4).

MM starts from the opposite end

MM starts from the other end. Its foundation is the non-local phase. The local world is the same reality read from a point.

If so, non-locality cannot appear only in entanglement. It must be at work wherever something is in a place.

What this deposit does

Deposit 2.68 is a philosophical deposit. There the declared role of the non-local phase was to give a reason for two facts:

- that the laws are the same everywhere;
- that the two ends of an entangled system are one nexus.

The *ends* are the two extremities of the system, for example the two distant particles. The *nexus* is MM's starting relation: we see it in §1.

The author's premises and the other deposits, however, use the non-local phase in many other places. This deposit puts them in a row. For each one it separates four things:

- what is a *measured datum*;
- what is a *known theorem*, that is, a result already proved;
- what is a *registered calculation*, that is, a calculation fixed in a deposit, with the rules written beforehand;
- what is a *premise*.

And it says where the link with the non-local phase comes from. It may already be in a deposit. It may come from a premise of the author. Or it is proposed here by the AI.

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1. What "non-local" means in MM

Not at a distance, but nowhere

In MM "non-local" means **not located in space and time**. It does not mean "at a distance". Nor does it mean "faster than light".

Space and time are not the container of the relation: they are not a box that holds it. They are the way it is read.

Two phases, but not two stages

The framework distinguishes two phases: a non-local phase and a local phase. They are not two stages, one before and one after. Nor are they two floors of a building, one above the other.

- The non-local phase has no time. So it comes before nothing.
- It has no place. So it is not in a location from which a distance separates us.
- It is present in every thing, in every place and at every instant.

The local phase is the same reality counted from a point.

A changed way of speaking. In earlier deposits the same thing is also called "level" or "underlying step". Here it is said without the image of floors: present everywhere. So does the *entry page* of the framework, that is, the deposit that presents MM.

The nexus between space and time

In the non-local phase there is only a logical distinction: space is not time.

A reminder of arithmetic is needed. The **greatest common divisor** (GCD) of two whole numbers is the largest number that divides both of them. For 3 and 5 the GCD is 1.

Space and time, the two distinct terms, have a single element in common. MM calls it their greatest common divisor, the

GCD. It is as with 3 and 5, which have only 1 in common.

Between 3 and 5 there is no intermediary. Nothing travels from one to the other. The relation holds entirely and at once.

Counting gives rise to the local world

The relation can be counted through its GCD. Then the *multiples* of the GCD arise (twice, three times, and so on) and its *fractions* (one half, one third, and so on). Multiples and fractions together form a **matrix**. Here the word does not mean a table from algebra: it means the set of the multiples and the fractions. The matrix is the local world.

An observer is inside the matrix. For the observer the multiples appear as time and the fractions as space.

Two postulates

A *postulate* is a starting statement: it is not proved, it is assessed by its consequences. Two postulates of the framework, from the entry page (record 22667516), do the work in everything that follows.

- **Existing is reciprocal:** an entity exists for another only to the extent of what they have in common. An *entity* is any thing at all.
- **Copies have no distance:** three copies are the same unit counted three times, and between one copy and another there is no distance. Think of 3: it is 1 counted three times. The three copies of 1 are the same unit.

Hence the sentence that sums up the framework: **the local is the non-local counted**. Every distance is made of copies between which there is no distance.

The four senses of the word

In philosophy of physics "non-local" has at least four senses. 2.68 separates them as follows.

- **Non-location.** The relation is not in a place or at an instant. *In MM:* it is the sense of MM.
- **Non-separability** (Howard 1985; Teller 1986). Two systems do not have separate states of their own. The *state* of a system is its complete description in physics. *In MM:* it is a consequence. Two copies of the same unit are not two things.
- **Bell non-locality** (Bell 1964). These are correlations that no explanation reproduces, if the explanation uses only properties of their own, influences no faster than light and independent measurements. *In MM:* it is a datum. MM reads it with non-separability and with the *loop* (§3.1).
- **Action at a distance.** An influence that goes from here to there faster than light. *In MM:* it is excluded.

What is not claimed. The non-local phase transmits no signals and no influences. Experiments have *events* (what happens), *settings* (the choices on how to measure) and *records* (the written results). All this is in *spacetime* as always (2.48, §3). Spacetime is space and time taken together, as in relativity.

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2. One mechanism, three ways of seeing it

Explanatory premise (AI). The organisation in three ways is the AI's; the premises are the author's. The author accepted it on 28 October 2026.

An *explanatory premise (AI)* is a proposal of the AI to put the author's ideas in order.

In the twelve phenomena of §3 the non-local phase always works in the same way. But it is seen from three sides. We call them way A, way B and way C.

Way A: copies without distance

What appears in many places is the same unit counted several times. And there is no distance between the copies.

From this follows a likeness at a distance that no message carried. Here are four examples:

- the two ends of an entangled system;

- two places with the same laws;
- two observers who see the same chair;
- the same limit of light for everyone.

Way B: existence passes through a link

An entity exists for another to the extent of what they have in common. And what they have in common is a non-local link.

Where the link is missing there is no hidden object. For that observer there is nothing.

For the same reason nothing can exist outside the web of relations. Not even other universes.

It is the hardest side of the framework. It is at work in observed objects and in dark matter.

Way C: the reading is local and partial

For the author it is the non-local basis that forces the present to a part of the matrix. The non-local link "switches on" only the part to which the observer belongs.

A *reading* is what an observer obtains from its own matrix. In MM an observer is any part of the structure that records and compares: not necessarily a person.

The reading that follows is local and partial. And some limits that we measure belong to it:

- the Tsirelson bound, which "is set by the local reading" (§3.1);
- the present (§3.8);
- the threshold beyond which an object stops being at a single point (§3.7);
- the boundary within which a system reads only its own field (§3.12).

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3. The twelve phenomena

Each phenomenon has six entries:

- **What is observed.**
- **Known physics:** how today's physics reads it.
- **What the non-local phase does** in MM's reading.
- **Where the link comes from.**
- **What rests on calculations** already deposited.
- **What remains open.**

3.1 The correlations of entanglement and their ceiling (ways A and C)

First, a few words.

- A *photon* is a particle of light.
- *Spin* is a property of particles that is measured along a chosen direction. For an electron the measurement always gives one of two results, called up and down. Spin is not a little ball that spins.
- In Bell experiments a quantity called **S** is measured. S puts together the correlations obtained with two different settings for each side. The Bell limit, for S, is 2.
- The number after the ± sign is the *uncertainty* of the measurement. It says how far, with good probability, the true value can be from the value given.
- A *loophole* is a way in which an ordinary explanation could still work. For example: the detectors lose too many particles; or the two sides are close enough to exchange a signal during the measurement. The best experiments *close* the main loopholes, that is, they exclude them.

What is observed. Correlations beyond the Bell limit are measured, with the main loopholes closed:

- two electrons in laboratories 1.3 km apart (Hensen et al. 2015, $S = 2.42 \pm 0.20$);

- atoms 398 m apart (Rosenfeld et al. 2017, S = 2.221 ± 0.033).

Other experiments also go beyond the limit:

- photons sent from a satellite to stations 1,203 km apart (Yin et al. 2017, S = 2.37 ± 0.09);
- tests in which the settings are chosen by the light of stars and quasars (Handsteiner et al. 2017; Rauch et al. 2018). A *quasar* is the very bright centre of a very distant galaxy.

In these three experiments (Yin, Handsteiner, Rauch) a *fair sampling* of the detectors is assumed. It means that the particles detected are an honest sample of all those produced.

The correlations cannot be used to send messages. And they do not exceed a second, higher limit: the **Tsirelson bound**. It is $2\sqrt{2}$, that is, 2 times the square root of 2. The experiment of Poh et al. (2015) obtained 2.82759 ± 0.00051. The central value is 0.00084 below $2\sqrt{2} = 2.82843$: very close to the ceiling, but below.

In 2022 the Nobel Prize in physics went to Aspect, Clauser and Zeilinger for experiments with entangled photons.

Known physics. *Quantum mechanics* is the theory that describes atoms, electrons and light. It predicts these correlations with the entangled state. The ceiling comes from a theorem: Tsirelson's theorem (1980).

The *interpretations* of quantum mechanics are different ways of saying what happens beneath the calculations, which stay the same. The interpretations differ on what produces the correlations.

What the non-local phase does. For MM the two ends are not two systems with properties of their own. They are two copies of the same nexus (way A).

Let us go back to Bell's three assumptions, in the form used here. MM drops the first one at the root: *separability*, that is, the idea that each system carries its own properties with it. For MM this is a premise of the framework, not a result of the theorem.

The ceiling, instead, is set by the local reading of each end (way C).

Where the link comes from. From deposits 2.48 and 2.68 (§1 and §5), and from the overview (ch. 4). The *overview* is an overall view of MM.

What rests on calculations (MM 2.48 and 2.48.1).

- *The ceiling.* Suppose that each end is read with the Born geometry, that is, with the rules of quantum mechanics for probabilities (the Born rule, §3.6). Then no correlation exceeds $2\sqrt{2}$. It is a theorem, already in the scientific literature.
- *A common record.* Suppose instead that the two ends carry only a common record, written at departure, like the gloves in the boxes. Then the correlations would stay below 2. But experiments exceed 2.
- *The singlet.* The *singlet* is an entangled state of two particles with spin. If the spin of the two particles is measured along the same direction, the results are always opposite. Along different directions, how often they are opposite depends on the angle between the directions: it is the *angular law*. The calculation uses *projective* measurements (the ordinary measurements of quantum mechanics) and *coplanar* ones (with all the directions in the same plane). And it assumes the angular law.

With these conditions the correlations of the singlet are reproduced in the *loop model*, with every link inside the light cones. The *light cone* of an event includes all the points of spacetime linked to that event by signals at the speed of light or slower. In the model the correlations pass through the common past of the two readings. Nothing goes directly from one end to the other.

The *loop* is the closure of a history between preparation and outcome, in which the future contributes to choosing the past. Close ideas are those of Aharonov, Bergmann and Lebowitz (1964), of Cramer (1986), of Argaman (2010) and of Wharton and Argaman (2020).

The price is Bell's third assumption. The choices of measurement are not independent of the state, because the future enters its preparation.

What remains open.

- Non-separability alone does not give the numbers of the correlations. The Born reading at the two ends gives them.
- Experiments exclude Bell's three assumptions taken together. They do not choose MM among the interpretations.
- *Relativistic causality* is the rule of relativity by which no influence leaves the light cones. Whether the model respects it still has to be argued separately (2.48, §5).
- Whether the non-local phase and the loop are the same fact seen from two sides is a reading of the AI (2.68, §5).

3.2 The same laws everywhere and always (way A)

What is observed. No difference has ever been measured between the laws in different places and epochs.

A precise check concerns one constant: the **fine-structure constant**, written α (alpha). It is a number without units of measurement. It fixes the strength of the electric and magnetic forces between charged particles.

- *Far away in space and time.* The *spectrum* of a light is the light divided by colours. Atoms leave lines in it at precise colours, and their position depends also on α. The number z says how much the light of a distant object has been stretched by the expansion of the universe: the larger z is, the longer ago the light set out. In the spectrum of a quasar at z = 1.15, α varies by no more than a few parts per million (Murphy et al. 2022). The result is $\Delta\alpha/\alpha = 1.3 \pm 1.3 \pm 0.4 \times 10^{-6}$. Δα/α is the change of α divided by α itself; the result has two uncertainties; 10^{-6} means one millionth.
- *Today, in the laboratory.* In today's atomic clocks α varies by no more than about 10^{-18} per year, that is, a billionth of a billionth (Lange et al. 2021).

Known physics. For physics the *universality* of the laws, that is, the fact that they are the same everywhere and always, is a starting assumption.

There is a famous theorem, **Noether's theorem**. It ties *symmetries* to the quantities that are *conserved*, that is, that stay constant:

- if the laws do not change when one shifts in time, energy is conserved;
- if the laws do not change when one shifts in space, *momentum* (mass times velocity) is conserved.

A shift of this kind is called a *translation*. Not changing is called *invariance*. The theorem ties symmetries to conservation laws. But it does not explain why the symmetries hold.

Physics assumes universality and checks it, without giving it a reason.

What the non-local phase does. Every "here" and every "now" is a copy of the same nexus. And between two copies of the same unit there is no distance.

Two places cannot have different laws for the same reason that two copies of 1 are not two different units.

The universality of the laws is non-locality seen from inside the local.

The laws do not need a guardian to make them respected. 3 and 5 do not "obey" their GCD: they are that fact.

Where the link comes from. From deposits 2.68 and 2.68.1 (§7) and from the overview (ch. 6).

What rests on calculations. No calculation: it is an argument, that is, a line of reasoning. It concerns the laws. The single values come from measurement: MM calls them *history*.

What remains open. Suppose that a different law were measured in a distant region. Then this reading would lose its support.

3.3 Identical particles (way A)

Proposal of the AI, not contained in earlier deposits; the author accepted it on 28 October 2026.

What is observed. All electrons have the same mass and the same charge. And they are *indistinguishable* in the strict sense: if two of them are exchanged, no prediction changes. An example chosen by us, for comparison: two identical

billiard balls can always be told apart, for example by following where each one goes. Two electrons cannot.

Electrons belong to a family of particles, the *fermions*. For fermions the state changes sign under exchange: in the mathematical description of the pair a minus appears. Which sign holds for which species of particle is fixed by a theorem, the *spin-statistics theorem*.

Known physics. *Field theory* is the language of particle physics. A *field* is a quantity present at every point of space. In this theory every electron is an *excitation* of the same field, that is, a small packet of energy of that field. Why there is a field for each species of particle, the theory does not ask.

The physicist Richard Feynman recounts an episode in his Nobel lecture (1965). The physicist John Wheeler told him on the telephone that all electrons have the same charge and the same mass "because they are all the same electron".

In philosophy it is debated whether two identical particles are two objects. Saunders (2003) argues that they are, in a weak sense.

What the non-local phase does. Two electrons are two copies of the same unit, counted twice. They are indistinguishable because between the copies there is nothing that makes them different.

The reading is close to that of field theory: the common field is the common nexus.

Where the link comes from. It is a proposal of the AI.

What rests on calculations. No calculation. The step from the copies to the sign of exchange has not been made.

What remains open. It remains open why different copies give different species of particles. For example the electron and the *muon*, a particle with the same charge as the electron but heavier. Without this step the argument would prove too much. An argument *proves too much* when, if it were right, it would also give conclusions that the facts contradict.

3.4 The objects that two observers see alike (ways A and B)

What is observed. Two people look at the same chair. The light that reaches their eyes is made of different photons. Yet they agree on where the chair is.

In laboratories a piece of information becomes *objective*, that is, the same for everyone, in a precise way. Many copies of the information spread into the environment. Whoever reads a small part of them finds the same value. It is **quantum Darwinism** (Zurek 2009). The name recalls Darwin and natural selection.

Known physics. Quantum Darwinism accounts for the agreement with the redundancy of the copies: there are many more of them than needed.

Then there is **Wigner's friend**, a thought experiment. An observer, the friend, makes a measurement inside a closed laboratory. A second observer, outside, treats the whole laboratory, friend included, as a quantum system.

The theorems on Wigner's friend (Bong et al. 2020) show one thing. Suppose that quantum mechanics holds also for an observer controllable as a quantum system. Then three things cannot hold together:

- the *absoluteness of outcomes*: every measurement has a single result, which holds for everyone;
- *locality*: no influence faster than light;
- the *absence of superdeterminism*: the choices of measurement are not tied in advance to the state being measured.

What the non-local phase does. The *record* of an observer is what it has collected. The two records of the two observers are two counts of the same non-local relation, made from two different points. Their common part is the trace of that relation (way A).

The observed object arises as a *non-local transposition and projection* of the observer's matrix. These are words of the framework. The observer projects its own matrix and discards the private part, that is, the part that only it has. In this way it isolates the trace of the relation. And that trace has no distance between its copies.

It is the same idea with which the framework reads entanglement. The two distant measurements exchange nothing.

They are the two ends of a single relation.

And there is a prohibition: two observers who have enough of the object cannot disagree.

Then the hardest side. What has no non-local link with us casts no shadow in our record. And for us it does not exist (way B).

A separate case is a laboratory that lets no copies out. For this case MM gives up the absoluteness of the outcome (2.48, §5).

Where the link comes from. From the deposits (2.62.1, §2.2 and §2.5) and from the author's premises of 12, 13 and 23 October.

What rests on calculations (MM 2.62 and 2.62.1). The calculations use a model of quantum Darwinism. There the common object is the *Gács–Körner common part* of the two records. It is a mathematical measure of the information that two records have in common exactly: each of the two can obtain it alone, without talking to the other.

The *bit* is the unit of information: the information of a yes-or-no answer.

- With a tolerance of 10^{-6}, that is, accepting a deviation of one millionth, 23 copies per observer are enough. Then the common part is almost the whole object: about 1 bit, almost all of the question "where is the chair?".
- The exact common part, without tolerance, is zero until the records are compared.
- With an object that can have more values, the common part grows *in steps*, that is, in jumps.

What remains open.

- The calculations test the common part, not the non-local phase.
- Identifying the copies of the framework with *scattered photons*, that is, photons that bounced off the object, is a choice of the AI.
- Agreement between the readings is already guaranteed by standard quantum mechanics. MM recovers it, but does not predict it in its own right.

3.5 The boundary of the local is a count (ways A and C)

First, a few words. An *interferometer* is an apparatus that splits a wave over two paths and then joins them again. Where the two parts reinforce or cancel each other, light and dark stripes appear: the interference *fringes*. In the quantum world atoms and molecules also give fringes. The *wavelength* is the distance between two successive crests of a wave.

What is observed.

- *A single photon is enough.* In an atom interferometer the two paths can be about half a wavelength of light apart. Then a single scattered photon, that is, one that bounced off the atom, erases the fringes. At larger distances the fringes come back in part, at intervals (Kokorowski et al. 2001).
- *Heat is enough. Fullerenes* are carbon molecules shaped like a ball. Heated fullerenes lose their fringes: in the measurements they go from 47 % to 29 %, to 7 %, to 0 % (Hackermüller et al. 2004). The cause is the *thermal photons* that fly away, that is, the light that every hot body emits.
- **The copies carried away count even if nobody collects them.**

Known physics. Known physics calls this phenomenon **decoherence**. It is the passage to *classical* behaviour, that of everyday objects. It depends on two things:

- how many copies the environment carries away;
- the separation between the paths compared with the wavelength.

It does not depend on the distance from whoever records.

What the non-local phase does. If the local is the non-local counted, its boundary cannot be a distance from the observer. It is a number of copies.

Let us take an ordinary record: a surface of 1 cm² at 300 K. The *kelvin* (K) is the unit of absolute temperature; 300 K is about the temperature of a room.

- In one microsecond, that is, a millionth of a second, this record leaves about 4×10^{12} copies of itself: four thousand billion.
- For a complete reading a few tens are enough: 21 for 99 % of the information. In the calculation the copies overlap by 0.9, that is, they are imperfect copies, which on their own distinguish little.
- At zero copies no local reading exists.

Where the link comes from. From a status report of August, the status report 2.9-2.18 (document QUANTUM_PLACEMENT, §0). Here it is reread in the *pre-geometric* sense: the sense of a relation that comes before the geometry of space.

A correction. That August text used "non-local" in Bell's sense, that is, for the correlations of entanglement. 2.48 corrected it: in MM the word has the sense of §1, not located in space and time.

What rests on calculations.

- The count of copies of the status report 2.9-2.18. The same number, 21 copies for a 1 % *deficit*, appears in 2.62.1. A 1 % deficit means that 1 % of the information is missing.
- Reading D1 of 2.66, compatible with standard decoherence. 2.66 is the deposit on the mass threshold (§3.7); D1 is its reading for the state of an object.

What remains open.

- The step "matrix → place" has not been made. That is, it is not derived which correlations follow from the count.
- Agreement between readings with enough copies is guaranteed by standard quantum mechanics. MM recovers it.

3.6 Probability (way C)

What is observed. Take a plate with two *slits*, that is, two thin openings. Behind the plate there is a screen.

- Send a single particle towards the slits. It arrives at a single point of the screen.
- Now send many of them, one after another. Point after point, the fringes of a wave are drawn on the screen.

Each single arrival is unpredictable. But the probability of each point follows a rule. It is the square of a number called the *amplitude*, more precisely of its size (we see this further below). This is the **Born rule**.

An example with numbers chosen by us: if the amplitude of an outcome is 0.5, its probability is $0.5 \times 0.5 = 0.25$, that is, one case in four.

Known physics. In known physics the Born rule is a *postulate*: it is assumed, not derived.

Some have tried to derive it by counting *branches* or copies (Zurek 2005). A branch is one of the possible outcomes, seen as a history of its own. These derivations rest on a contested step: equal copies, equal probabilities (Schlosshauer and Fine 2005).

What the non-local phase does. For the author the non-local phase and the *mirror reciprocity* between the two ends of the nexus give a reason for the probabilistic behaviour of matter.

The framework's image is this. One has to describe a 10 having only the 1. Then the 1 must be distributed over ten positions, one for each unit of the 10.

Where the link comes from. From the author's premise of 22 October. It is in the box "MM in brief", the short summary that appears on all MM records.

What rests on calculations (MM 2.40, 2.48 and 2.48.1). Some words are needed. ψ (read psi) is the *wave function*: the quantum description of a system, which gives the amplitudes. Amplitudes are *complex* numbers: besides a size they have a *phase*, a kind of direction, like an arrow. The *modulus* of an amplitude is its size, that is, the length of the arrow. $|\psi|^2$ is the square of the modulus: the probability of the Born rule.

- *Counting the copies.* Counting copies of equal modulus gives $|\psi|^2$. Three assumptions are needed, and they are

declared. The result holds in the cases tested: the ratios with denominator up to 12, that is, fractions such as one third or seven twelfths. To these is added *continuity* for irrational ratios, that is, for the numbers that are not fractions.

- *The exponents.* The calculation also tested other exponents, that is, other powers in place of the square. Among these, the step of the *checkerboard* of 2.40 conserves only the sum of squares. The checkerboard is a model of 2.40 in which a particle advances in steps along the directions of light.
- *The loop.* In the loop the Born weight is already contained in the closure rule. For the Born rule, the loop reading is *circular*: it uses what it would like to derive. And the *measure over futures*, that is, the weight given to each possible future, remains a postulate.
- *Self-consistency.* Then there is an argument of *self-consistency* (2.48, R3). Self-consistency means that a history does not contradict itself. The argument holds in the family of protocols tested: a pair model, with the angular law cos² assumed (cos² is the square of the cosine of the angle). The argument looks at the consistent histories with weights other than Born's. Their weight becomes as small as one likes as the number of pairs grows. But it stays positive for every finite number of pairs. The impossibility of these histories is not proved. And the step to "they do not exist" is a postulate of the author.

What remains open. Two facts are not derived: that probabilities exist, and which outcome occurs.

3.7 When an object stops being at a single point (way C)

First, a few words. A *superposition* is a state that combines two or more conditions, for example two positions. It is not an object in two places at once: it is a single state, which can give interference fringes. The *atomic mass unit* is about the mass of a proton. A *nanoparticle* is a very small grain of matter.

What is observed. Today nanoparticles of more than 7,000 atoms and 170,000 atomic mass units also give interference fringes (Pedalino et al. 2026).

For everyday objects, instead, under ordinary conditions no interference between distinct positions is observed.

No experiment has yet found the boundary.

Known physics. Decoherence accounts for the classical look of large bodies (§3.5).

Collapse models, instead, look for a true limit, present even in a vacuum, without an environment. They are modifications of quantum mechanics in which a superposition ends by itself. An example is the model of Diósi and Penrose, which ties the limit to gravity.

What the non-local phase does. When the object is large enough, its mass fixes it in the matrix of the point of observation. Then we see it in one place and at one instant.

Below a certain level of mass things change. Two GCDs become equal, as in a mirror:

- the GCD of the point of observation;
- the GCD of the *diffraction* that generates the object. Here diffraction is a word of the framework: it indicates how the observed object arises.

Then the object no longer occupies a single point of the observer's spacetime. It spreads over several positions.

In the framework's reading the boundary is the point at which the object becomes the common part of the records. It is the bridge with §3.4.

Where the link comes from. From 2.66 and from the author's premise of 22 October: «il sistema logico a specchio e la fase non locale giustificano il comportamento probabilistico della materia» ("the logical mirror system and the non-local phase account for the probabilistic behaviour of matter").

What rests on calculations (MM 2.66).

- *The readings.* 2.66 wrote ten readings of the threshold as a mass and eight readings of the distance. They were registered before the calculations: written and fixed before calculating, so they could not be adjusted afterwards.
- *The two readings that hold.* Two readings of the distance hold. One concerns the *state* of the object (D1). The other concerns the *reading*, that is, how the object appears to the observer (D6g). They coincide with decoherence and with *wave optics*, the part of physics that studies light as a wave. The framework admits them;

it does not predict them differently.
- *The clock.* The reading with the clock of 2.40 describes a condition of the apparatus, that is, of the reading. It does not describe a property of the object.
- *The prediction.* A prediction is registered with a date: 31 December 2035. It comes with a program that anyone can apply. The most likely outcome in 2035 is "not assessable", because the planned experiments do not reach the effect.

What remains open.

- The number for the threshold.
- Whether D1 and D6g are a single rule of the framework. It is a proposal of 2.66.

3.8 The present and the choice of what is written (way C)

What is observed. Every measurement gives a single outcome. And what is written stays written.

Usually heat goes from the hot body to the cold one. An everyday example: a cup of tea left on the table cools down; it does not warm up by itself. In some experiments, however, with initial correlations between the parts, heat goes from cold to hot (Micadei et al. 2019).

Known physics. In the equations of physics the present has no place: there is no special instant called now.

The *arrow of time* is the fact that time has a direction, from the past to the future. Known physics ties it to heat and to *initial conditions*, that is, to how a system started.

What the non-local phase does. For the author the non-local basis forces the so-called present to a part of the matrix: the part to which the observer belongs. The non-local link "switches on" only one part of the total matrix.

Also for the author, which reading is written is decided by the future of the history, which chooses its past. It is the loop.

The non-local phase has no before or after. In the AI's reading the loop is how that unity appears in the local reading.

Where the link comes from. From the author's premises of 11 and 13 October, published in 2.48 and 2.68. The translation of the loop into a model in light cones is the AI's (2.48).

What rests on calculations (MM 2.48, 2.64.1 and 2.68.1). 2.68.1 made a *single calculation*: loop, record and direction of time stand in one model.

The calculation uses the rule in which the future chooses by the closure of the loop. With this rule three things happen:

- the first readings have the same statistics whether the history stops or continues;
- the record grows and gives the *Born frequencies*, that is, those predicted by the Born rule;
- in the *Micadei testbed*, the test model built on the Micadei experiment, the record grows while heat flows backwards. It happens on average and along the most probable history.

On the checkerboard of 2.40, instead, the record grows less than the threshold set beforehand: 0.29 and 0.17 bits, against 0.5. And the condition "only the past", on its own, does not select the Born rule.

What remains open. What distinguishes the "switched-on" part of the matrix from the part that is switched off (2.68, §10).

3.9 Space and time themselves (way A)

What is observed. For someone at rest, a moving clock runs slower: Einstein's relativity says so. The time marked by a clock along its path is called *proper time*. Proper time is a fact measured with atomic clocks.

Known physics. In known physics spacetime is the starting stage: everything happens inside it.

Quantum gravity is the search for a theory that unites gravity and quantum mechanics. There one looks instead for a spacetime that *emerges*, that is, that arises from something that is not spacetime (Huggett and Wüthrich 2025). For some, the *connectedness* of space, that is, the fact that its regions are connected, comes from entanglement (Van Raamsdonk

2010).

What the non-local phase does. For the author a non-local layer has in itself the features to generate space and objects.

If the local is the non-local counted, places are not given beforehand. They arise with the count.

A clock that counts events without assigned places must give the time we measure. And places given beforehand must act as a *privileged frame*, that is, as a special point of view.

Where the link comes from. From the author's premise of 10 October. The link with the calculations of 2.40 is the AI's, because 2.40 speaks of events without places, not of the non-local phase.

What rests on calculations (MM 2.40 and 2.40.2). Some words are needed.

- The *arithmetic mean* of two numbers is their sum divided by two. The *geometric mean* is the square root of their product. An example with numbers chosen by us: for 4 and 9 the arithmetic mean is 6.5; the geometric mean is $\sqrt{36} = 6$.
- The *two light sides*. Between two ticks of a clock one can draw a diamond made of light rays: one to the right and one to the left. The two sides of the diamond, measured along the directions of light, are the two light sides.
- The factor $\sqrt{(1 - v^2)}$ says how much a moving clock slows down: it is the factor of *time dilation*. Many books call its inverse, $\gamma = 1/\sqrt{(1 - v^2)}$, the *Lorentz factor*: the two express the same dilation, one multiplies and the other divides. Here v is the velocity, measured as a fraction of the speed of light. An example with numbers chosen by us: at $v = 0.8$ the factor is $\sqrt{(1 - 0.64)} = 0.6$. While 10 seconds pass for someone at rest, the moving clock marks 6.
- Hermann Bondi was a physicist who presented relativity with signals of light sent and received.

Here are the results.

- *Einstein's time.* A clock that reads the number of copies runs at the geometric mean of the two light sides. That is, it runs with Einstein's proper time. Grant the premise of counts on the two sides (as in Bondi). Then the factor $\sqrt{(1 - v^2)}$ is an *identity*: it always holds, given how things are defined.
- *Without assigned places.* The copies are placed at random. One takes the *longest chain*: the longest row of events, each one after the previous one. The chain recovers the factor within about 1 %, in 1+1 dimensions (one of space and one of time).
- *On a grid of places.* The same reading gives the arithmetic mean: Galileo's time, that is, the time of physics before Einstein.
- *With fixed places and random times.* The dilation is 7 to 29 % stronger than Lorentz's, because the places define a frame.
- *In 3+1 dimensions*, that is, three of space and one of time. With about 6,000 copies, clocks with the same proper time are compared. The difference is within 1.3 σ. The symbol σ (sigma) measures statistical uncertainty: a difference of about one σ is of the kind that chance often produces.
- *At high speed.* At $v = 0.8$, with about 780 copies, the deviation exceeds the fixed 6 % threshold. That threshold had been registered beforehand, for the group of tests called block P1-P5.
- *More than two sides.* One can also count on more than two sides. Then only the *pairwise rule* keeps the Lorentz factor. And n sides give $n - 1$ directions.

A close idea is that of Knuth and Bahreyni (2014).

What remains open.

- The number of dimensions is not derived.
- The *pre-geometric programme* is the project of making the geometry of space arise from the framework, instead of starting from it. Gravity, *Lorentz invariance* and the 3+1 dimensions are not derived from the pre-geometric programme alone (2.30). Lorentz invariance is the fact that the laws are the same for all observers in uniform motion, as relativity requires.

3.10 The limit of light (way A)

What is observed. No body and no signal locally exceeds the speed of light in vacuum. This speed is the same for all observers.

Known physics. In relativity it is a starting principle. Since 1983 the metre has been defined from it: it is the distance

that light travels in vacuum in a fixed fraction of a second.

What the non-local phase does. For MM the speed of light is the *fixed rate of exchange* between the multiples and the fractions of the same GCD. It says how much space one step of time is worth. A comparison chosen by us: it is like the fixed exchange rate between two currencies, which says how many units of one are worth one unit of the other.

Exceeding it would mean, for some observer, covering space without counting time. That is, leaving the local world.

In the AI's reading the rate is the same for everyone because every observer counts copies of the same nexus, as for the laws of §3.2.

Outside the system there is no other term to refer it to.

Where the link comes from. The fixed rate, and exceeding it as leaving the local world, are in two places: in the box "MM in brief", approved by the author, and in 2.68, §7. The link with the copies of the same nexus is the AI's.

What rests on calculations. No calculation. The closest calculation is the geometric mean of 2.40 (§3.9).

What remains open. Why the rate has precisely the measured value. Inside the system it is a *necessary relative*: a relative value, which in there holds for everyone. Outside, it has no justification.

3.11 Dark matter as a second mirror (way B)

What is observed. Galaxies and *clusters of galaxies* (large groups of galaxies bound by gravity) move as if there were about five times more matter than the matter that emits or absorbs light. This matter that cannot be seen is called *dark matter*.

Planck is a satellite that measured the oldest light in the universe. The *standard cosmological model* is the model most used to describe the universe. In this model the Planck data give a *cold* and *non-baryonic* component. It is about 5.4 times the density of baryons (Planck Collaboration 2020).

- *Baryons* are the particles of ordinary matter, such as protons and neutrons.
- *Cold* means that its particles move slowly.

Known physics. One proposal is a new particle, which gravitates and does not interact with light. No candidate particle has been confirmed.

There are also models in which dark matter is ordinary matter of a *mirror sector*: a set of particles similar to ours, as in a mirror, with forces of their own (Kolb, Seckel and Turner 1985; Berezhiani 2004; Foot 2004).

What the non-local phase does. For the author dark matter is ordinary matter of another mirror.

- A large part of it is real for its own local GCD, to which it is bound non-locally.
- It is not bound non-locally to the observer's GCD. So, in the principal mirror, it is non-existent.
- There is a reciprocal non-existence, for lack of a reciprocal non-local link.

Yet it gravitates. For the author the *fractionation* acts on the electromagnetic interaction, not on the gravitational one. Fractionation is a word of the framework: the dividing of the count into fractions. The *electromagnetic interaction* is the force between electric charges; light is also part of it.

In way B, then, "it does not exist for us" holds for light and for electromagnetic interactions. It does not hold for gravity.

For the author, finally, what is decisive in the difference between the two mirrors is the non-local share of exchange within each.

Where the link comes from. From the author's premises reported in 2.44 (words 2, 5, 8 and 12).

What rests on calculations (MM 2.44).

- *A hidden sector.* 2.44 describes a hidden sector with its own light.
- *A numerical hypothesis.* It uses the hypothesis $\Omega_c/\Omega_b = 3\sqrt{3}$, supplied separately by MM. Ω_c is the amount

of cold dark matter, Ω_b that of baryons. 3√3 means three times the square root of 3.

- *The cost.* Imposing this hypothesis costs T_B = 0.17-0.23 with the data of Planck and of DESI DR2. T_B measures how much the agreement with the data gets worse when the ratio is imposed, instead of being left free. DESI is a large survey of galaxies that measures the history of the expansion; DR2 is its second data release. The result is *compatible*, in an *a posteriori* check, that is, one made after seeing the data. A test on future data is registered.
- *The RAR.* The *radial acceleration relation* (RAR) ties the acceleration measured in galaxies, towards their centre, to the one predicted by visible matter alone. The author's words on the share of exchange motivated a form of the RAR written with L/S, a ratio between two quantities of the framework. It is not a derivation.
- *One more hypothesis.* The reading also postulates that the hidden radiation, that is, the light of the hidden sector, does not source our expansion.
- *A limit.* The worked example does not establish a *viable* hidden sector: it does not show that such a sector can really work.

What remains open. The relative constants of the second mirror are left in a generic form. A complete calculation of the hidden sector is missing.

3.12 The boundary of a gravitating system, set by the Galaxy (way C)

First, a few words.

- A *wide binary star* is a pair of stars that orbit each other at a great distance.
- An *acceleration* says how much a velocity changes in each second.
- **MOND** (modified Newtonian dynamics) is a theory that changes Newton's law when the accelerations are very small.
- The *internal dynamics* of a system is the way its parts move relative to each other.

What is observed. A wide binary star, near the Sun, sits in the gravitational field of the Galaxy, the Milky Way.

At low accelerations galaxies follow a common scale, a_0 (read a-zero). In MOND the internal dynamics of a system also depends on the external field it sits in.

Known physics. In Einstein's gravity the *strong equivalence principle* holds. It says that a *uniform* external field, that is, one that is equal at every point, does not change the internal dynamics of a system. An example chosen by us: in a lift in free fall, objects stay still relative to one another, as if there were no gravity.

In MOND this principle does not hold.

What the non-local phase does. What a system reads depends on the context it belongs to, not on the point.

For the author the behaviour depends on the *observational context*. The Galaxy acts on the observation GCD. And the observation GCD, in turn, determines the system's own field.

In the AI's reading, here the non-local shows as context. The law of a system depends on the whole, as in a reading and not as in a force.

Where the link comes from. From the author's premises of 27 and 28 October. In 2.70 the hypothesis "a non-local element of context is needed" is said to agree with two premises: that of 10 October (the non-local layer) and that of 27 October. It is consistency, not proof.

What rests on calculations (MM 2.70).

- *The local versions do not hold.* 2.70 tried to join MM's dynamics, that is, its equations of motion, with the a_0 scale. The *local* versions of this junction tried so far do not hold. Some do not pass the comparison with Cassini, the space probe that studied Saturn: its data check gravity in the Solar System with great precision. These versions are 13.7 to 23.5 σ beyond the limit. A distance of many σ is a clear disagreement, which chance practically does not produce. Others do not pass the granularity test, another check of 2.70.
- *The hypothesis of the context.* The hypothesis of the non-local context remains motivated, but not verified. And the tests do not show that it is necessary.
- *The translation T_max.* The AI translated the premise of 28 October into a rule, called T_max: "the boost with which a system is read does not exceed the boost with which the Galaxy reads the point". The *boost* is how many times the attraction read exceeds the one predicted by Newton's law. With T_max two things come back

that before were only *posited*, that is, assumed without deriving them. The first is the *boundary*, where the system's own field falls to the level of the external one. The second is the *isotropy* inside the boundary: the behaviour is the same in all directions.

- *The Gaia pairs.* Gaia is a satellite that measures positions and motions of stars; DR3 is its third data release. On the Gaia DR3 pairs, T_max is the best of eight hypotheses. But the comparison is a posteriori. And T_max does not separate from *MM's chain*, another reading of MM for the same pairs.
- *A globular cluster.* A *globular cluster* is a compact ball of very many stars. The *parsec* (pc) is a unit of distance for stars, a little more than three light years. For the globular cluster NGC 2419 the law gives a boost of 3.3-3.4 at 86 pc. The data instead give at most about 1.1 ($\lesssim 1.1$). Here the law does not agree with the data, like every law made of the internal and external accelerations alone. This outcome strikes the law T_max, the AI's translation. The proposed way out is the history of the cluster, that is, how it formed and changed over time. The premise does not replace this history.
- *A registered rule.* The rule for Gaia DR4 is registered.

What remains open.

- The translation has two declared choices: the *scalar* reading (a single number, without direction) and the *maximum* (the boost of the system does not exceed that of the Galaxy). Why precisely the maximum is not derived.
- With DR4 the most likely outcome is "not decidable". It holds both for the amplitude of the effect and for its direction. An *observable* is a quantity that can be measured. With the direction observable (Part A of 2.70) about ten times the DR3 pairs would be needed to reach 2σ.

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4. Summary table

The table puts the twelve phenomena in a row. The *origin* column says where the link between the non-local phase and the phenomenon comes from:

- **D**: already written in a deposit;
- **A**: premise of the author;
- **AI**: proposed by the AI.

The translations into models and calculations are always the AI's. They are stated in the calculations entry of each phenomenon (for example §3.4, §3.8, §3.12).

phenomenon	way	what the non-local phase does	origin	what rests on calculations	deposits
correlations of entanglement	A, C	the two ends are one nexus; the ceiling is set by the reading	D	Tsirelson with local Born; loop for the singlet	2.48, 2.48.1
same laws everywhere	A	every here and every now is a copy of the same nexus	D	argument	2.68, 2.68.1
identical particles	A	two electrons are the same unit counted twice	AI	none	—
shared objects	A, B	two records count the same relation; without a link nothing exists for us	D, A	common part in a quantum-Darwinism model	2.62, 2.62.1
boundary of the local	A, C	the boundary is a number of copies	D (reread)	count of copies; decoherence	status report 2.9-2.18, 2.62.1, 2.66
probability	C	describing the 10 with the 1 forces one to distribute it	A	Born from counting with three assumptions	2.40, 2.48, 2.48.1
quantum threshold	C	below a mass the reading that fixes gives way	A, D	registered readings; prediction for 2035	2.66
present and writing of outcomes	C	the non-local link switches on part	A, D	single calculation of loop, record and direction	2.48, 2.64.1, 2.68.1

		of the matrix; the loop chooses			
space and time	A	places arise with the count	A	proper time from events without places	2.30, 2.40, 2.40.2
limit of light	A	the same rate for whoever counts copies of the same nexus	D, AI	none	box "MM in brief", 2.68
dark matter	B	without a reciprocal non-local link there is no common light	A	hidden sector; 3√3 imposed, compatible a posteriori	2.44
boundary of gravitating systems	C	the Galaxy fixes the reading of the system	A, D	local versions beyond the Cassini or granularity limits; T_max on DR3 a posteriori; DR4 test registered	2.70

What the table says. The twelve phenomena are far apart from each other: they go from the scale of photons to that of the Galaxy. In all of them MM's reading uses the same premise in the same way.

- In ten phenomena the link is already in the deposits or in the author's premises.
- In one it comes partly from the AI: the limit of light.
- In one it comes from the AI alone: identical particles.

The result of the deposit is this unity of the reading. In MM's reading non-locality is not an exception of entanglement. It is the common ground of facts that known physics accounts for with separate assumptions.

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5. What is not claimed

- The non-local phase transmits no signals and no influences outside the light cones.
- Today the non-local phase has no *distinct experimental signature*, that is, a laboratory result that only it predicts. Where MM gives numbers, there are two cases:
- the numbers coincide with those of known physics: the Tsirelson bound, decoherence, wave optics;
- or they come from readings with declared choices (T_max, a posteriori).
- The readings are not the only possible ones. The same observations have other interpretations, and experiments do not choose among them.
- The deposit does not solve the *measurement problem*: the question of how, in a measurement, among many possible outcomes only one occurs. And it does not derive the Born rule without assumptions.
- The translations of the premises into calculations are the AI's. They are declared as such in the deposits cited.

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6. Where the reading is put to the test

- **Gaia DR4 (December 2026).** In 2.70 the rules for the boundary set by the Galaxy are registered (§3.12). The date fixes the test. The most likely outcome is "not decidable".
- **The mass threshold (31 December 2035).** It is the dated prediction of 2.66. Anyone can apply it with the deposited program (§3.7).
- **Experiments that change only the distance from whoever records.** There are none today (2.66, §4.6). They would make the role of distance testable (§3.7).
- **Five open questions** (2.68, §10):
- if in the non-local phase there is no measure and no before or after, what makes the original GCD begin to be counted?
- what distinguishes the "switched-on" part of the matrix from the part that is switched off?
- in what sense is the reading of a record non-local?
- in entanglement, what relation is there between the non-local phase and the loop?
- if the non-local phase is a real fact, what is the main physical consequence that today's physics sees but does not explain?

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7. One word, several senses: "non-local" elsewhere in the deposits

In MM's records the word "non-local" also appears in different senses. These senses must not be confused with the non-local phase.

- **Covariant memory** (August status report, §7). In Einstein's gravity, gravity is the *curvature* of spacetime. Here there is a factor that depends on the *integral* of the curvature, that is, on its sum, over the whole past light cone. This factor is from the family of *non-local gravity* cited in that text. And there is a memory along the *trajectory*, that is, the path, of the body. It is non-locality in the sense of field theory. The August text itself says that the two uses coincide "by linguistic accident".
- **The horizon coupling** $r_t = \sqrt{(R_s \cdot R_H)}$ (status report 2.9). R_s is the *Schwarzschild radius* of a body, the radius at which it would become a black hole. R_H is a length tied to the *cosmic horizon*, that is, to the scale of the whole universe. r_t is a local scale tied to the cosmic horizon. Later MM adopted $a_0(z)$, that is, the value of a_0 at the different epochs of the universe. The $a_0(z)$ adopted later uses the elapsed-age clock, that is, the age of the universe used as a clock. This clock is local.
- **Matter in compact stars** (2.34.2). *Compact stars* are very dense stars. Here ψ is not the wave function of §3.6: it is a quantity of the framework used in that deposit. Through ψ the *stress* at a point, that is, the pressure and the internal forces of matter, depends on the whole configuration.
- **The reading operator on the lattice** (2.52). A *lattice* is a grid of points used for calculations in place of continuous space. Here the word concerns locality in the sense of *lattice fermions*, that is, of particles such as the electron described on that grid.
- **The law L*** (2.70). In 2.70 it is the reading in which a pair of stars, inside a boundary, reads only its own field. It is "non-local" because it depends on the whole system and not on the point. It is tied to the non-local phase only through the author's premises, as consistency (§3.12).

A correction. The "Bell" sense used in the August texts on the *quantum placement*, that is, on the place of quantum mechanics in the framework (status report 2.7-2.27), was also corrected in 2.48. In MM "non-local" means not located in space and time.

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8. The neighbours

The *neighbours* are ideas of other authors similar to those of this deposit.

- **The implicate order** of Bohm (1980): what appears separate is enfolded in an undivided whole.
- **Non-separability** as a thesis distinct from locality: Howard (1985), who reads it in Einstein; Teller (1986), with *relational holism*, the idea that a whole has relations that the parts alone do not have.
- **Emergent spacetime:** Huggett and Wüthrich (2025). The connectedness of space from entanglement: Van Raamsdonk (2010). Spacetime from a network of influences: Knuth and Bahreyni (2014).
- **Bell non-locality and relativity:** Maudlin (2011).
- **Retrocausal readings with local mediation:** Aharonov, Bergmann and Lebowitz (1964); Cramer (1986); Argaman (2010); Wharton and Argaman (2020). *Retrocausal* means that the future also contributes to determining the past. *With local mediation* means that every link passes from a point to a nearby point.
- **Objectivity from copies:** Zurek (2009). *Relational quantum mechanics*, in which the properties of a system hold relative to another system: Rovelli (1996).
- **Identical particles:** Feynman (1965) on Wheeler's idea; Saunders (2003) on *weak discernibility*, that is, on a weak way of telling two particles apart.
- **Dark matter as matter of a mirror sector:** Kolb, Seckel and Turner (1985); Berezhiani (2004); Foot (2004).

How the work was done

In the archived text this part corresponds to section 9 and to the section Contents of the package.

Who did what. The premise of the non-local phase and the sentences from which the readings start are the author's, Ugo Lanciano, who is not a physicist. His exact words, with the dates, are in the file AUTHOR_BACKGROUND.md. The translations of the premises into calculations, the calculations and the texts of the deposits cited were made with the AI (Claude). Other systems checked them (Astra, Gemini); the checks are described in each deposit.

How the archived text was written. The text was written by the AI in Italian on 28 October 2026, at the author's request. It started from the published records, read in the local copies identical to the deposited files. The English text is

its translation: it is the file README.md.

The reviews. A *remark* is a critical observation by whoever checks.

- A second instance of the same AI, that is, a separate session, compared the first draft with the texts of the deposits. It made 21 remarks, all checked again against the sources and applied (REVIEWS/REVIEW_1_CLAUDE_IT.md).
- The author read the second draft and accepted it, including the two proposals of the AI (§2 and §3.3).
- Then two external systems, Astra (Codex) and Gemini (Argon), checked the final texts in two rounds against extracts of the deposits cited.
- In the first round: 17 remarks, almost all applied, and one rejected on the merits with the reason.
- In the second round Gemini answered "ready" and Astra "ready with corrections", with three remarks applied.
- The reports are in REVIEWS/REVIEW_2_ASTRA_ARGON_G1_IT.md and REVIEWS/REVIEW_3_ASTRA_ARGON_G2_IT.md.
- The new references were checked on open pages (REFERENCE_CHECK.md).

One precise source. The count of copies of §3.5 is the program plateau_copie.py of the status report 2.9-2.18.

Contents of the package.

file	what
README.md	the text in English
LEGGIMI_IT.md	the Italian text (the original)
AUTHOR_BACKGROUND.md	the author's sentences on the non-local phase, with exact words and dates
REFERENCE_CHECK.md	how the new references were checked
REVIEWS/	the reports of the checks on the text
LICENSE.md	licence (CC BY 4.0)
IMPRONTE.sha256	SHA-256 fingerprints of the files

A SHA-256 *fingerprint* is a code calculated from the content of a file: if the file changes even slightly, the fingerprint changes.

There is no code in the deposit. The calculations it cites are in the deposits listed in the section Deposits cited.

This version. This student version was written with an AI assistant (Claude) from the archived Italian text (LEGGIMI_IT.md). The numbers were checked with a program against the archived text. The archived text is the reference.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

The words of the framework (MM)

- **AI's translation:** the way in which a sentence of the author becomes a quantity that can be calculated. It is a choice. An outcome that depends on a translation concerns that translation.
- **Box "MM in brief":** the short summary that appears on all MM records.
- **Checkerboard:** a model of 2.40 in which a particle advances in steps along the directions of light.
- **Copy:** the same unit counted once more. Between two copies there is no distance.
- **Entry page:** the deposit that presents the framework (2.27).
- **Explanatory premise (AI):** a proposal of the AI to put the author's ideas in order.

- **Fractionation:** the dividing of the count into fractions. For the author it acts on the electromagnetic interaction, not on gravity (§3.11).
- **GCD:** the common element of the two terms of a nexus. In arithmetic it is the greatest common divisor: for 3 and 5 it is 1.
- **Local phase, matrix:** the same reality counted from a point. The multiples of the GCD appear as time, the fractions as space.
- **Loop:** the closure of a history between preparation and outcome, in which the future contributes to choosing the past.
- **Nexus:** the relation between two distinct terms that have a single element in common, as 3 and 5.
- **Non-local (in MM):** not located in space and time. It does not mean at a distance or faster than light.
- **Non-local phase:** reality as pure relations, without space, without time and without objects. It is present everywhere, neither before nor beneath the rest.
- **Pre-geometric programme:** the project of making the geometry of space arise from the framework, instead of starting from it.
- **Premise:** a starting sentence of the author, a fixed point like a postulate.
- **Reading:** what an observer obtains from its own matrix. The observer is any part of the structure that records and compares.
- **Record:** what an observer has collected.
- **Ways A, B, C:** the three sides from which the non-local phase is seen (§2). A: copies without distance. B: existence passes through a link. C: the reading is local and partial.

The words of physics

- **a_0:** the common scale of acceleration that galaxies follow at low accelerations.
- **A posteriori:** said of a check made after seeing the data.
- **Acceleration:** how much a velocity changes in each second.
- **Amplitude:** the number that quantum mechanics gives to each outcome. It is a complex number, with a size (the modulus) and a phase. The square of its modulus is the probability.
- **Baryons:** the particles of ordinary matter, such as protons and neutrons.
- **Bell limit:** the limit that the three assumptions of Bell's theorem, together, put on correlations. For the quantity S it is 2.
- **Bit:** the unit of information, that of a yes-or-no answer.
- **Boost:** how many times the attraction read exceeds the one predicted by Newton's law.
- **Born rule:** the probability of an outcome is the square of the modulus of its amplitude, $|\psi|^2$.
- **Correlation:** the agreement between two results, more frequent than chance would make it.
- **Decoherence:** the loss of interference fringes when the environment carries away copies of where an object is.
- **Entanglement:** a state of two or more particles, even distant ones, that must be described together.
- **Fermions:** the family of particles to which electrons belong. When two fermions are exchanged the state changes sign.
- **Field:** a quantity present at every point of space. In field theory each species of particle has its own field.
- **Gács–Körner common part:** the information that two records have in common exactly; each can obtain it alone.
- **Globular cluster:** a compact ball of very many stars.
- **Interference fringes:** the light and dark stripes that appear where two parts of a wave reinforce or cancel each other.
- **Interferometer:** an apparatus that splits a wave over two paths and then joins them again.
- **Light cone:** the set of the points of spacetime linked to an event by signals at the speed of light or slower.
- **Loophole:** a way in which an ordinary explanation could still account for the results of a Bell experiment.
- **MOND:** modified Newtonian dynamics, a theory that changes Newton's law at very small accelerations.
- **Noether's theorem:** the theorem that ties the symmetries of the laws to the quantities that are conserved.
- **Parsec (pc):** a unit of distance for stars, a little more than three light years.
- **Photon:** a particle of light.
- **Proper time:** the time marked by a clock along its path.
- **Quantum Darwinism:** the idea that a piece of information becomes objective when many copies of it spread into the environment (Zurek 2009).
- **Sigma (σ):** the measure of statistical uncertainty.
- **Singlet:** an entangled state of two particles with spin. Along the same direction the two measurements always give opposite results.
- **Spin:** a property of particles that is measured along a chosen direction. It is not the rotation of a little ball.
- **Strong equivalence principle:** in Einstein's gravity, a uniform external field does not change the internal dynamics of a system.
- **Superdeterminism:** the idea that the choices of measurement are tied in advance to the state being measured.
- **Superposition:** a state that combines two or more conditions, for example two positions. It is not an object in two places at once.

- **Tsirelson bound:** the highest value, 2√2, that the correlations of quantum mechanics can reach in the quantity S (Tsirelson's theorem, 1980).
- **Wave function (ψ):** the quantum description of a system, which gives the amplitudes.
- **Wigner's friend:** a thought experiment. An observer measures inside a closed laboratory; a second observer, outside, treats the whole laboratory as a quantum system.
- **z:** how much the light of a distant object has been stretched by the expansion of the universe.

Names of instruments and data

- **Cassini:** the space probe that studied Saturn; its data check gravity in the Solar System.
- **DESI DR2:** the second data release of DESI, a large survey of galaxies that measures the history of the expansion.
- **Gaia DR3:** the third data release of Gaia, the satellite that measures positions and motions of stars.
- **Planck:** the satellite that measured the oldest light in the universe.

Deposits cited

record	what	DOI
status report 2.7-2.27	the August quantum placement, the count of copies	10.5281/zenodo.21802475 (concept)
entry page 2.27	the non-local phase present everywhere	10.5281/zenodo.22667516
2.30	the pre-geometric programme	10.5281/zenodo.22730712
2.40, 2.40.1, 2.40.2	clock without places, checkerboard, pairwise rule	10.5281/zenodo.23280167 (concept); 2.40.2: 10.5281/zenodo.23631239
2.44	dark matter as a second mirror	10.5281/zenodo.23329080
2.48	Tsirelson bound, loop	10.5281/zenodo.23352814
2.48.1	Born from counting copies	10.5281/zenodo.23655646
2.62, 2.62.1	objects as common part	10.5281/zenodo.23609966 (concept); 2.62.1: 10.5281/zenodo.23643717
2.64.1	record and direction of time	10.5281/zenodo.23643718
2.66	mass threshold, distance	10.5281/zenodo.23643759
2.68, 2.68.1	what "non-local" means, single calculation	10.5281/zenodo.23669889 (concept); 2.68.1: 10.5281/zenodo.23672394
2.70	boundary set by the Galaxy, DR4 test	10.5281/zenodo.23730218
overview	overall view	10.5281/zenodo.23535916 (concept)

References

Each reference is linked to the open page on which it was checked: in deposits 2.44, 2.48, 2.62.1, 2.66 and 2.68 for those already used there, and on 28 October 2026 for the new ones (REFERENCE_CHECK.md).

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